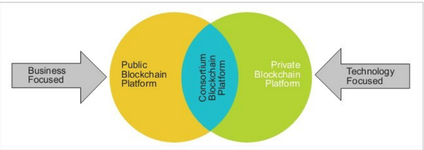

Figure 1: Blockchain consortium

Note: The figures in per cent indicate the percentage of the 1386 enterprises that said a blockchain consortium had benefited them with respect to the parameter indicated.

How a blockchain consortium differs from public and private blockchain platforms

Blockchain is based on a peer-to-peer topology that allows data to be stored globally on millions of servers. It is also described as a P2P, decentralised, distributed ledger technology that works without involving any third-party intermediary or central authority, unlike traditional banks that rely on intermediaries. Blockchain offers security, transparency and immutability, and is attracting developers because of its potential to revolutionise the way industries and businesses work.

As per a Deloitte survey in 2019, at least 76 per cent of enterprises who are using blockchain solutions are moving towards a blockchain consortium or willing to move to it in the next 2-5 years due to its agility in architecture. Blockchain platforms are classified as private, public, semi-private and consortium. In fact, semi-private blockchain is the early version of a hybrid blockchain platform as it offers an enterprise level public blockchain platform for businesses.

As stated earlier, depending on their use and requirements, blockchains have been categorised into three types —public, private and consortium (also known as federated). Each blockchain solves particular problems, and has its own set of features and advantages.

Public blockchain

This blockchain is a permissionless, non-restrictive, distributed ledger system, which means anyone who is connected to the Internet can join and become a part of it. The basic use of such a blockchain is for exchanging cryptocurrencies and mining. Moreover, it maintains trust among the whole community of users as everyone in the network feels incentivised to work towards the improvement of the public network. The first example of such a blockchain is Bitcoin, which enabled everyone to perform transactions. Litcoin and Ethereum are also examples of a public blockchain.

Private blockchain

Unlike the public blockchain, a private blockchain is a permissioned and restrictive blockchain that operates in a closed network. It is mostly used in an organisation where only a few members are participants of the blockchain network. It is best suited for enterprises and businesses that want to use blockchain only for internal purposes. The major difference between the private and public blockchains is that the latter is highly accessible, whereas the former is confined to a particular group of people. Moreover, a private blockchain is more centralised because a single authority maintains the network. Corda, Hyperledger Fabric and Hyperledger Sawtooth are examples of private blockchains.

Consortium blockchain

A consortium blockchain (also called federated blockchain) is best suited for organisations where there is a need for both types of blockchains, i.e., public and private. In this type, there is more than one organisation involved in providing access to pre-selected nodes for reading, writing and auditing the blockchain. Since there is no single authority in control, it maintains its decentralised nature. Energy Web Foundation and IBM Food Trust are examples of such a blockchain.

Table 1 highlights the differences between public, private and consortium blockchains.

Types of blockchain consortia

There are three types of blockchain consortia:

- Technology-focused
- Business-focused
- Dual-focused

Technology-focused: A technology-focused blockchain consortium offers reusable blockchain platforms, and solutions based on technical standards. It has multi-purpose use cases. This type of consortium exists solely for the purpose of helping blockchain achieve global recognition. Examples are Quorum, R3 Corda and Hyperledger.

Business-focused: This is intended to develop blockchain solutions for a specific business issue. Instead of offering open source platforms, many prefer these consortia for commercial purposes only. Most of these are focused on industries like supply chain, health, trading, healthcare, etc. Examples are Bankchain, We.Trade, B3i, Marco Polo, etc.

Dual-focused: The focus here is both on technology and business when offering a platform or solution. These offer open source platforms suitable for the solution and commercial products as well. An example is R3.